

§15 Multiple Integrals

§15.1 Double Integrals over Rectangles

Homework : 3,11,13,17,18

Problem : $z = f(x, y) \leftrightarrow$ 房子的屋頂 ($f \geq 0$)

$D = [a, b] \times [c, d] \leftrightarrow$ 房子的地基 ($f \geq 0$)

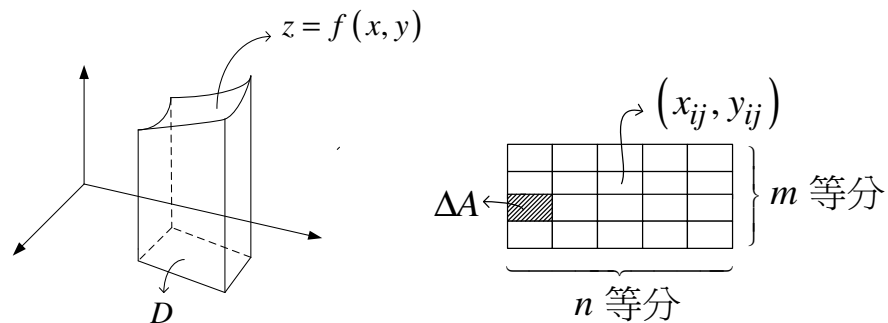
= a rectangle

則此房子的體積(Volume)為何?

Definition : The double integral of f over a rectangle D is

$$\lim_{n \rightarrow \infty} \sum_{i=1}^m \underbrace{\sum_{j=1}^n f(x_{ij}, y_{ij})}_{\text{高}} \underbrace{\Delta A}_{\text{底}} = \text{volum under the roof } z = f(x, y) \text{ on } D$$

$D(f \geq 0) =$ 若 $f \geq 0$, 此式代表屋子的體積.



Remarks :

(i) $\sum_{i=1}^m \sum_{j=1}^n f(x_{ij}, y_{ij}) \Delta A$ is called double Riemann sum

= 一堆小長方形之體積和

= 這是二維的算法

(ii) Midpoint Rule : Double Riemann sum with (x_{ij}, y_{ij}) being center of the corresponding rectangle.

(iii) average value of f on $D = \frac{\iint_D f(x, y) dA}{\text{area } D} = \text{房子的平均高度}.$

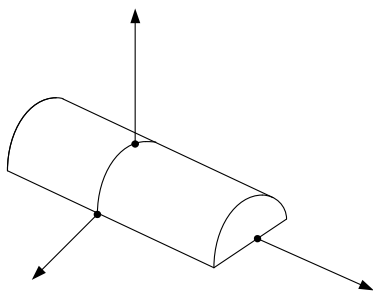
(iv) Properties of double integral.

(v) linear : $\iint_D (af + bg) dA = a \iint_D f dA + b \iint_D g dA$

(vi) $\iint_D f \geq \iint_D g$ provided that $f \geq g$ on D .

Example 1 : 屋頂 : $f(x, y) = \sqrt{1-x^2}$

地基 = $D = [-1, 1] \times [-2, 2]$



Solution :

Evaluate $\iint_D \sqrt{1-x^2} dA = \text{圓柱體的一半}$

$$= \frac{1}{2} \pi (1)^2 \cdot 4 = 2\pi$$

Example 2 : $z = x^2 + y^2$. $D = [0, 1] \times [0, 1]$. Using midpoint rule to compute the

double Riemann sum with $n = m = 2$.

$$\iint_D (x^2 + y^2) dA \approx \frac{1}{4} \left[f\left(\frac{1}{4}, \frac{1}{4}\right) + f\left(\frac{3}{4}, \frac{1}{4}\right) + f\left(\frac{1}{4}, \frac{3}{4}\right) + f\left(\frac{3}{4}, \frac{3}{4}\right) \right] = ?$$

